C PROGRAMMING SYLLABUS



Module 1: Introduction to C Programming

- What is C? History, features, applications, and program structure
- Character set, tokens, keywords, identifiers, constants, and variables
- Data types overview; compiling, linking, and executing a C program
- Compiler/IDE setup and first C program
- Types of errors: syntax, runtime, and logical errors
- Algorithms, flowcharts, and pseudocode

Module 2: Data Types, Operators, and Expressions

- Basic data types, type modifiers, and ranges
- Variables, constants, type conversion, and type casting
- Arithmetic, relational, logical, assignment, increment/decrement, bitwise, shift, and conditional operators
- Operator precedence and associativity
- sizeof operator and format specifiers

Module 3: Decision-Making and Loop Control

- if, if-else, nested if, else-if ladder, and switch
- while, do-while, and for loops; nested loops
- break, continue, and brief introduction to goto with its limitations
- Menu-driven programs, loop tracing, and debugging
- Practical programs: even/odd, largest of numbers, prime, palindrome, Armstrong, perfect number, factorial, Fibonacci, and patterns

Module 4: Functions and Recursion

- Function declaration, definition, prototype, function calls, and return values
- Parameters and arguments; call by value
- Using pointers to modify values through functions
- Local and global variables; scope and lifetime
- Recursion and standard library functions
- Modular programming and code reuse
- Practical programs: recursive factorial, GCD, prime checking using functions, and calculator

Module 5: Arrays, Searching, and Sorting

- One-dimensional, two-dimensional, and multidimensional arrays
- Passing arrays to functions; traversal, insertion, and deletion
- Linear search and binary search
- Bubble sort and selection sort
- Array of pointers and array memory layout
- Practical programs: maximum/minimum, average, reverse array, duplicate detection, matrix addition, and transpose

Module 6: Strings

- Character arrays, string storage, and the null character (\0)
- String input/output and safe input with buffer-size awareness
- Pointers and strings
- String library functions: strlen, strcpy, strcat, and strcmp
- Character handling using ctype.h and manual string processing
- Practical programs: reverse string, palindrome, vowel/word counting, string comparison, and character frequency

Module 7: Pointers and Dynamic Memory Allocation

- Pointer declaration, address operator (&), and dereference operator (*)
- Pointer arithmetic; relationship between arrays and pointers
- Pointers with functions and pointer-to-pointer
- Null, dangling, and uninitialized pointers
- malloc, calloc, realloc, and free
- Memory leaks and invalid memory access
- Practical programs: swap using pointers, dynamically allocate an array, resize memory, and process array elements using pointers

Module 8: Structures, Unions, and User-Defined Types

- Structure declaration, initialization, and member access
- Array of structures and nested structures
- Pointers to structures and the arrow operator (->)
- Passing structures to functions
- Unions, enum, and typedef
- Practical programs: student, employee, and book records

Module 9: Storage Classes and Preprocessor

- Scope, lifetime, and linkage
- auto, register, static, and extern storage classes
- Introduction to the C preprocessor and its role before compilation
- #include and Header Files
- Macros using #define and #undef
- #include, object-like and function-like macros
- Macro expansion and conditional compilation
- Practical: reusable macros and separating declarations into header files

Module 10: Console Input/Output and File Handling

- printf, scanf, character/string input-output functions, and format specifiers
- Return values and basic input validation
- FILE pointer; opening and closing files; file modes
- fgetc, fputc, fgets, fputs, fprintf, and fscanf

- Text and binary files; fread and fwrite
- File error checking; fseek, ftell, and rewind
- Practical programs: create/read/append text files, store employee records, and search records in a file

MINI-PROJECTS

- **Student Record Management System:** Store, display, search, update, and delete student records using structures and files.
- **Library Record Management System:** Maintain book details, search records, and track issue/return information.
- **Employee Management System:** Store employee information and perform record search and update operations.
- **Inventory Management System:** Maintain item names, quantities, prices, and stock information.
- **Expense Tracker:** Record expenses, calculate totals, and save/read entries from a file.

Note: Prepare and solve weekly multiple-choice questions (MCQs) based on the topics covered during the week.

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